Risshab Srinivas Ramesh

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LinkedIn

Research interests: Learning-based Combinatorial Optimization, Reinforcement Learning under constraints, Chaos-theoretic methods for search, and explainability in safety-critical ML systems.

Education

Ramaiah Institute of Technology *Sep 2023 – Present*
B.E., Computer Science and Engineering, CGPA: 9.91/10

- **Relevant coursework:** Data Structures and Algorithms, Operating Systems, Computer Networks, Deep Learning, Generative AI, DevOps

Indian Institute of Technology Madras *Sep 2023 – Present*
B.Sc., Programming and Data Science, CGPA: 8.8/10

- **Relevant coursework:** Machine Learning, Deep Learning, Probability and Statistics, Linear Algebra, Optimization Methods, DBMS, Business Data Management

Research Experience

Research Intern *Robert Bosch Centre for Cyber Physical Systems, IISc* *Mar 2025 – Present*

- Advised by Dr. Prasant Misra and Dr. Nithin Nagaraj
- Formulated the Electric Vehicle Routing Problem (EVRP) with time windows and energy constraints as a Markov Decision Process and trained a Deep Q-Network (DQN) agent to produce near-optimal routes.
- Integrated NeuroChaos Learning (NCL) as a chaotic feature map preceding the DQN policy network. NCL's ergodic exploration properties reduce the effective search horizon on large instances, yielding faster convergence than standard ε-greedy exploration.

Project Intern *Hewlett Packard Enterprise (HPE)* *2024*

- Designed and deployed a multi-agent log analysis system using LangGraph. An orchestrator agent dynamically routes log events to a specialised anomaly detection agent (embedding-based clustering over a VectorDB) and a root cause analysis agent (retrieval-augmented chain-of-thought reasoning), replacing manual triage.

Publications

Chaos-Inspired Ergodic Encoding for Combinatorial Optimization Current Research
Risshab Srinivas Ramesh, Dr. Prasant Misra, Dr. Nithin Nagaraj

- Introduced Ergodic Solution Encoding (ESE): rather than learning instance-level decision policies, ESE encodes the *population-level* distribution of optimal solutions as the invariant measure of a Piecewise Linear Markov Map (PLM), fitted via the Inverse Frobenius–Perron Problem.
- Solution generation uses Orbit Walk and Ergodic Beam Search, exploiting deterministic chaotic dynamics to produce diverse, high-quality candidates without stochastic sampling or instance-specific training.
- On 15 TSPLIB benchmarks (N=10–500): matches MILP optimality at small scale; outperforms GA and learning-to-optimize baselines at medium–large scale at orders-of-magnitude lower inference latency.

Learning-Based Optimization of EV Routing with V2G Integration and On-the-Go Energy Harvesting *COMSNETS 2026*
Risshab Srinivas Ramesh, Dr. Prasant Misra, Dr. Arjunan Pandaraswamy

- Formulated the EVRP as a reinforcement learning problem, jointly optimizing routing, V2G energy sales, and on-route solar harvesting. Modelled time windows and energy dynamics as MDP constraints.
- On the Solomon benchmark dataset, the learned agent runs **18.85× faster** than a genetic algorithm baseline; the GA achieves at most 21.23% higher solution accuracy, reflecting the practical speed-accuracy tradeoff at deployment scale.

When Explanations Fail: Measuring Attribution Drift in Corrupted Medical Images and an Architecture to Resist It *BiSCUIT @ ECCV 2026*
Risshab Srinivas Ramesh, et. al

- Introduced energy-in-mask (EiM), a chance-calibrated attribution metric that formalises a "sub-chance" test for saliency maps on segmentation models.
- Applied to CT lung nodules (LIDC-IDRI, n=150): the proposed ARX-SE-UNet, which conditions an SE-UNet backbone on a learned corruption embedding via FiLM-inspired blocks, is the only architecture whose EiM exceeds chance (0.0170 vs. $\bar{c}$=0.0122) and achieves **6.2**× the EiM recovery of the best baseline under CLAHE+BM3D preprocessing.

Optimizing Big Data Processing: Hadoop's Master-Slave Architecture and Serverless Alternatives *IEEE 2024*

Risshab Srinivas Ramesh, Pradeep Kumar

- Analysed Hadoop's master-slave scheduling under skewed workloads and characterised conditions under which serverless (FaaS) execution models offer superior cost-latency tradeoffs for batch analytics pipelines.

Projects

Smart MCQ Solver *GitHub*

- Benchmarked recurrent sequence models (LSTM, GRU) against fine-tuned pre-trained transformers (BERT, DeBERTa) on a domain-specific multiple-choice question dataset. Evaluated using MAP@3; DeBERTa fine-tuning outperformed all baselines, establishing the gap between in-context language modelling and task-specific recurrence for multi-choice reasoning.
- **Stack:** Python, PyTorch, HuggingFace Transformers

CNN for Dermatological Disease Detection *GitHub*

- Designed and trained a custom CNN architecture for multi-class classification of skin conditions from dermoscopy images. Systematic augmentation (rotation, colour jitter, cutout) and stratified cross-validation were used to address class imbalance and improve generalisation across under-represented skin types.
- **Stack:** Python, TensorFlow, Keras, OpenCV, Scikit-learn

Automated Code Generation and Deployment Pipeline *GitHub*

- Built a FastAPI service that converts natural-language specifications into complete, deployable codebases. An automated CI/CD pipeline containerises the output, pushes to GitHub, and deploys to GitHub Pages. Hosted on Hugging Face Spaces.
- **Stack:** Python, FastAPI, HuggingFace Transformers, Docker, Git

Technical Skills

Frameworks and Libraries: PyTorch, TensorFlow, LangChain, LangGraph, HuggingFace Transformers, FastAPI, Flask, NumPy, Scikit-learn

Programming Languages: Python, Java, C, SQL, JavaScript

Tools and Infrastructure: Docker, Git, Redis, Celery, VectorDBs (FAISS/Chroma), Linux

Databases: MySQL, SQLite, MongoDB

Honors and Awards

LRN Foundation Grant – Received a travel grant to attend ECCV 2026 *2026*

Best Blog Award – Society for Data Science Blogathon on Responsible AI *2024*

Smart India Hackathon – Winner, Tourism Category (national-level competition) *2022*

Karnataka Science and Technology Award – Finalist, among 100 statewide participants *2022*

Microsoft Learn Student Ambassador – selected to organise technical events and mentorship programmes on Microsoft platforms.

Certifications (selected): Goethe-Zertifikat B1 (German, Goethe-Institut); Google IT Automation with Python (Google); Python for Everybody Specialization (U. Michigan); C Programming with Linux (Dartmouth/IMT).